MCQA

Minecraft answers you can check

Cited Wiki evidence and official recipe calculations

Fully local · Minecraft Java 26.3 target

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Players need facts and exact amounts

Gameplay questions

Wiki pages mix editions and historical rules.

Retrieve passages and show revision links.

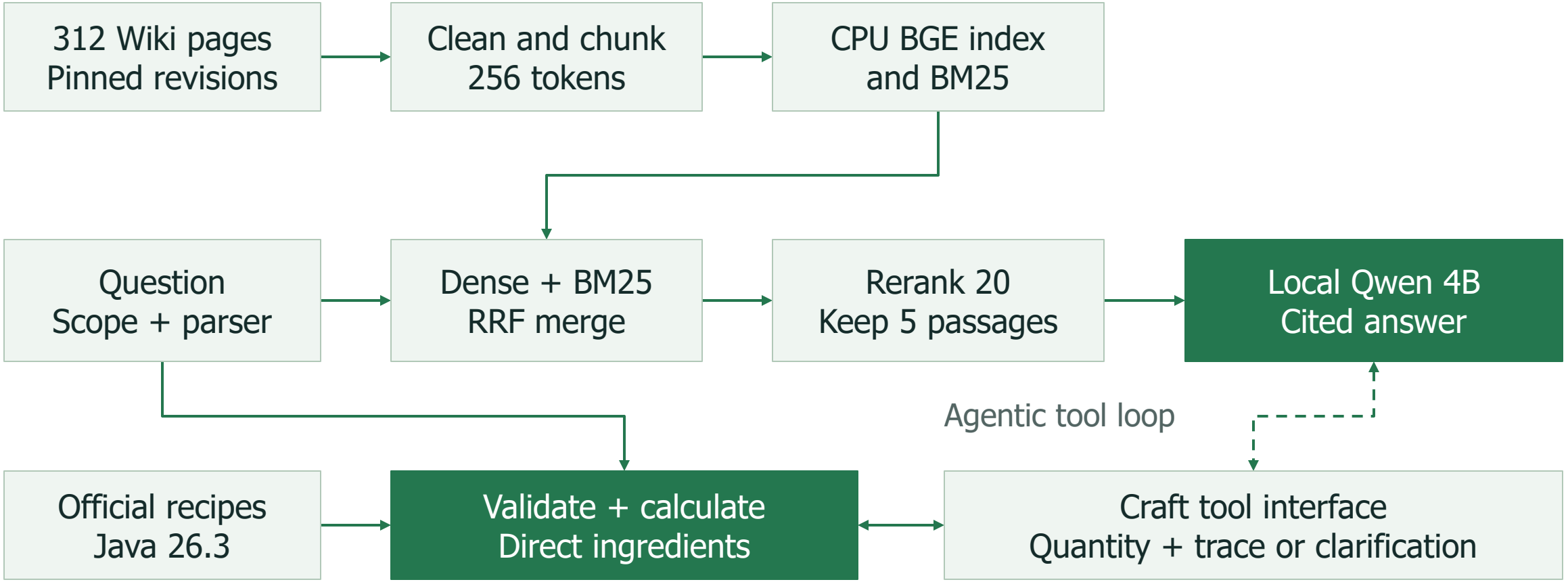
Crafting questions

"How many iron ingots for 3 hoppers?"

Read an official recipe and calculate the batches.

Scope: direct crafting and supported Java evidence. Other editions need a clear boundary.

Two answer paths, one local system



Clear recipes bypass generation

Embedding and reranking run on CPU. Generation uses local Ollama.

Corpus and chunking

312

distinct real Wiki pages

About 817,000 words
Exact page revisions retained

256-token chunks

Keep section and paragraph boundaries.
Split oversized paragraphs.
No overlap.

Compared 128, 256 and 512 tokens.
Historical reranker page hits
were similar: 74.1% to 74.5%.

The synthetic Q&A pool chooses page titles and supports older diagnostics. It is not the retrieval corpus.

Small models keep the system local

Role	Choice	Purpose
CPU embeddings	BGE-small-en-v1.5	384-dim semantic vectors
Hybrid search	BM25 + dense, RRF	Match names and meaning
CPU reranking	MS MARCO MiniLM-L6-v2	Joint passage scoring
Local generation	Qwen3 4B Instruct	Citations and tool calls

Retrieve 50 per branch. Fuse ranks. Rerank 20. Answer from the top 5.

Live demo next: recipe quantity, cited gameplay answer, edition boundary.

Retrieved evidence improves factual answers

Variant	Judged correct	Quote in top 5
No retrieval	9/34	Not applicable
Dense	27/34	31/34
Hybrid + reranker	26/34	30/34

Evidence gives the model facts it could otherwise guess.

Reranking did not establish a gain over dense retrieval.

Frozen v1 test: 34 facts + 6 negatives. Automatic same-model judge. Historical implementation.

The recipe tool improves quantity answers

Same 40 questions	Exact quantity	Correct (%)
Without recipe tool	1/40	2.5%
With recipe tool	37/40	92.5%

Official recipe slots determine ingredient consumption.

All 3 misses made no tool call, so routing is the weak point.

Frozen v2 test. Direct ingredients only, without inventory reuse or recursive material planning.

Both metrics favour retrieval

Variant	LLM judge correct	BERTScore F1
No retrieval	9/34	0.294
Dense	27/34	0.493
Hybrid + reranker	26/34	0.451

Correctness checks facts. BERTScore checks similarity.

Both compare with pinned references. Neither is human validation.

Same 34 test facts as slide 6. Same-model judge; reranking showed no gain over dense.

Correct numbers can still hide wrong claims

Observed failure	Why it happens
Useful Java evidence removed	Historical and edition rules share paragraphs
Right ingredient, wrong explanation	The model adds claims beyond the tool result
Judge verdict can be wrong	Generator and judge use the same local model

Keep raw answers, citations and tool traces for review.

Next: improve mixed-section handling and independently review labels on a fresh test set.

AI audit: three decisions and one caught error

Three AI-owned decisions

256-token section chunks: balance context and cost.

CPU BGE embeddings: compact English retrieval.

Official recipe calculation: exact quantities, with a tool loop.

One actual error catch

The old scorer credited a refusal that repeated a number.
Fix: check the requested ingredient and reject refusals.

Claude Code and Codex built and checked the system.